UIC Department of Chemical Engineering
COLLEGE OF ENGINEERING

ASSIGNMENT #2

Course:	CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term:	Spring 2020
Professor:	Dr. Michael Caracotsios
Office:	EIB 244
E-mail:	mcaracot@uic.edu

Your name: Sai Dereddy

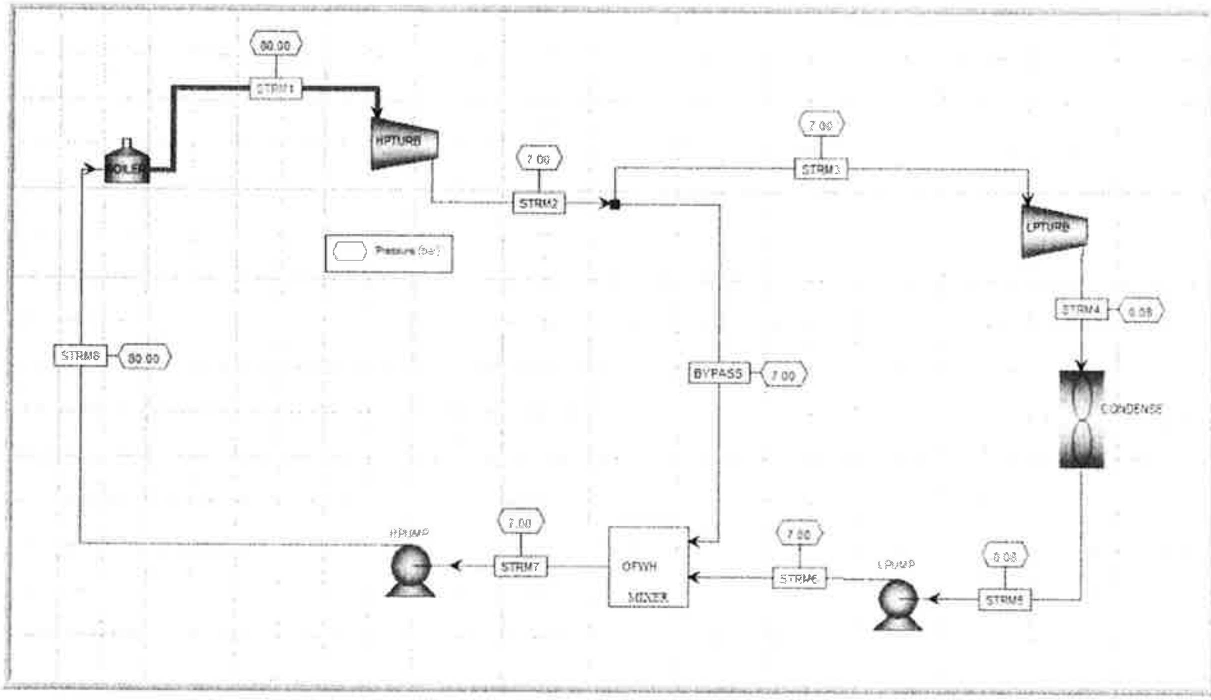
Due Date: Wednesday February 05 by 5:00pm in my office

Number of points.............................100

Your score...................................65

Note: All problems are worth 100 points. You final score will be the average value

1000 $MMSCFD$ of superheated steam generated in a boiler at $P_0 = 80\,bar$ and $T_0 = 480^{o}C$ are expanded into a steam condensing turbine (**CST**) to a pressure $P_2 = 8\,kPa$. To improve the efficiency of the power cycle a bypass stream is mixed with the discharge of the low pressure pump of the condenser effluent in order to preheat the boiler feed water. (See schematic below known as **Regenerative Rankine Cycle**).



If the turbine has an efficiency of 88% and the feed water pumps efficiency of 80% calculate:

1. The maximum flow rate of the steam bypass in **MMSCFD**
2. The amount of power generated by the cycle in **MW**
3. The efficiency of the cycle
4. The amount of water in **gpm** required for the condenser, assuming a water temperature rise of $11^{o}C$
5. If the power plant is using coal as fuel in the boiler give an estimate of the amount of coal needs in **tonne/day**
6. If the power plant is equipped with cooling tower, how much water does it draw from the environment, and how much it discharges as waste. Express your answers in **gpm**

To solve for the bypass we'll need to solve for each of the streams. To solve for each steam we'll need to solve the energy balance for each of the components.

[Turbine Energy Balance]

$$\Delta H = Q + W$$

[Turbine is an Adiabatic Process so Q =0]

$$\Delta H = W$$

[Condenser Energy Balance]

$$\Delta H = Q + W$$

[Condenser has no work, but there is a transfer of heat]

$$\Delta H = Q$$

[Pump Energy Balance]

$$\Delta H = Q + W$$

[Pump is an Adiabatic Process so Q =0]

$$\Delta H = W$$

Given that we are using 1000 MMSCFD of steam, we need to convert this into $\frac{\text{kmol}}{\text{s}}$

So the flowrate is 13.182 in kilomoles per a second. why? (-5)

Stream 1 Ideal: P = 80 bar, T = 480 and x =1

Units	S1
kmol/s	13.1620
kg/s	
m3/s	
C	480.0000
bar	80.0000
	1.0000
kg/kmol	18.0000
kJ/kmol	60302.4430
	60302.4430
	6029.4876
kJ/(kmol K)	119.9305
	119.9305
	0.0000

Given the initial conditions we can find the entropy of steam 1

Stream 2 = Steam 3 = Bypass Ideal: P = 7 bar

S2
13.1620
164.9560
7.0000
0.9908
18.0000
49380.9453
49723.7964
12549.1746
119.9305
119.5998
0.3307

Given the pressure is 7 bar, I solved for temperature using CHE 301 calculator. Also getting the enthalpy and entropy. Set the S2 entropy equal to the S1 entropy by changing the quality. I did this by using goal seek. OK

Stream 4 Ideal: P = .08 bar

S4
9.2568
41.5340
0.0800
0.7947
18.0000
37513.2114
46395.6262
3130.0892
119.9300
117.7399
2.1901

S3 to S4 is the Low Pressure Turbine in which that is a huge drop in pressure, as a result there should be a drop in temperature. Getting the temperature as 41.5340 degrees Celsius. Also calculating the quality of the vapor by knowing that the entropy in must equal the out.

[Entropy balance]

$$\Delta S = 0$$

Should use (-2)

$S_{3\,real} = S_{4\,ideal}$

[Entropy balance with numbers and lever rule]

$$6.6567 = 8.2231(x) + .5922(1 - x)$$

[Solving for x]

$$6.0625 = 7.6309x$$

[Getting x]

$$x = .79047$$

Stream 5 Ideal: p =.08

S4 to S5 goes thought a condenser which changes all of the quality to 0

S5
9.2568
41.5340 ← assumption
0.0800
0.0000
18.0000
3130.0892
46395.6262
3130.0892
10.6676
0.0000
10.6676

Stream 6 Ideal: P = 7 bar

The pump increases the pressure of water from .08 bar to 7 bar. Getting the results using the CHE 301 Calculator

S6
9.2568
41.5340
7.0000
0.0000
18.0000
3141.0375
49723.7964
3141.0375
35.8586
0.0000
35.8586

Stream 7 ideal: P =7 bar

Steam 7 is the mixing point of the bypass and steam 6:

13.1620
164.9560
7.0000
0.0000
18.0000
12549.1746
49723.7964
12549.1746
120.7131
0.0000
120.7131

Stream 8 Ideal: P =80

S7 to S8 is the pressure pump, only increasing the pressure and keeping temperature constant.

S8
13.1620
164.9560
80.0000
0.0000
18.0000
12624.5214
12624.5214
12624.5214
57.7467
0.0000
57.7467

Real Steams:

S1 Real	S2 Real	S3 Real	S4 Real	S5 Real	S6 Real	S7 Real	BYPASS	S8 Real
13.1820	13.1820	10.5225	10.5225	10.5225	10.5225	13.1820	2.6595	13.1820
480.0000	164.9560	164.9560	41.5340	41.5340	41.5340	164.9560	164.9560	164.9560
80.0000	7.0000	7.0000	0.0800	0.0800	7.0000	7.0000	7.0000	80.0000
1.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000	1.0000	0.0000
18.0000	18.0000	18.0000	18.0000	18.0000	18.0000	18.0000	18.0000	18.0000
60302.4430	50691.5250	50993.2340	38936.9948	3130.0892	3154.7228	12549.1746	50691.5250	12454.9911
60302.4430	50691.5250	50993.2340	38936.9948	46395.6262	49723.7964	49723.7964	50691.5250	12624.5214
6029.4876	12549.1746	12549.1746	3130.0892	3130.0892	12549.1746	12549.1746	12549.1746	12624.5214
0.0403								
					Qc	376779.2915		
					Qh	630725.1110		
turbine 1	-126691.1207	turbine 2	-126862.1573	pump 1	259.2078		pump 2	1241.5272
							total power	-252052.5429
							Efficiency	39.9623

Stream 1 Real: Steam 1 Real and Steam 1 ideal are the same

Stream 2 Real: I am the assumption that that steam coming out of the turbine is at saturated temperature which is 164.95 with the quality of 1. The enthalpy is calculated the efficiency of the turbine of 88%. So, the calculation the difference between the ideal enthalpy of S2 and S1 multiplied by the efficiency plus the enthalpy of S1.The work is the difference between S2 Real and S1 real $\left(\frac{\mathrm{KJ}}{\mathrm{Kmol}}\right)$ multiplied by the molar flow rate $\left(\frac{\mathrm{Kmol}}{\mathrm{s}}\right)$. Giving us the work $\left(\frac{\mathrm{KJ}}{\mathrm{s}}\right)$. Which would be -126691.12 $\left(\frac{\mathrm{KJ}}{\mathrm{s}}\right)$.

[Calculation for S2 Real]

Entropy or Enthalpy?

$$\text{S2 real} = (\text{S2 Ideal} - \text{S1 Ideal}) * 88 + \text{S1 Ideal}$$

[Work from the real High-pressure turbine]

$$(\text{S2} - \text{S1})\text{N}_2 = -126691.12\frac{\text{KJ}}{\text{s}}$$ (-2) X

Stream 3 Real: Steam 3 Real is the same as Steam 2 Real and Steam 3 Real has a quality of 1. Steam 3 has a different molar flowrate

Bypass Real: Same as Steam 2 Real except for the molar flow rate

Stream 4: The same assumption that steam coming out would be saturated and the quality is 1. The enthalpy is calculated the efficiency of the turbine of 88%. So, the calculation was the difference between the ideal enthalpy of S4 and S3 multiplied by the efficiency plus the enthalpy of S3.The work is the difference between S3 Real and S4 real $\left(\frac{KJ}{Kmol}\right)$ multiplied by the molar flow rate $\left(\frac{Kmol}{s}\right)$. Giving us the work $\left(\frac{KJ}{s}\right)$. Which would be -126862.16 $\left(\frac{KJ}{s}\right)$.

[Calculation for S4 Real]

$$\text{S4 real} = (\text{S4 Ideal} - \text{S3 Ideal}) * 88 + \text{S3 IDeal} \quad = ? \quad (-1)$$

[Work from the real Low-pressure turbine] should use H3 real (-2)

$$(\text{S4} - \text{S3})\text{N}_4 = -126862.16 \frac{\text{KJ}}{\text{s}} \quad (-2)$$

Stream 5: Turns the quality from 1 to 0 by removing heat

Stream 6: Low pressure pump would increase the pressure to 7 bar keeping temperature at
(-1) ? Psat. The Enthalpy was calculated with the efficiency of 80% of the pump. It was calculated by finding the difference in enthalpy between S6 Ideal and S5 ideal divided by the efficiency plus the enthalpy of S5 Ideal. The work was the difference in enthalpy from S6 and S5 multiplied by the molar flow rate.

[Calculation for S6 Real]

$$\text{S6 real} = \frac{(\text{S6 Ideal} - \text{S5 Ideal})}{.88} + \text{S5 Ideal} \quad = ? \quad (-1)$$

[Work from Real Low-Pressure pump]

$$(\text{S6} - \text{S5})\text{N}_6 = 259.20 \frac{\text{KJ}}{\text{s}} \quad (-2)$$

Steam 7: Is the mixing point of S6 and the Bypass. This is where we can calculate the molar flow rate of the bypass and S3 using an enthalpy balance.

1. To find the maximum flowrate in the bypass

[Enthalpy balance]

$$\Delta \mathrm{H} = \mathrm{W} = 0$$

[Enthalpy balance using flow rates]

$$H_7 * N_7 = H_6 * N_6 + H_B * N_B$$

[Mass balance]

$$\mathrm{N_7} = \mathrm{N_6} + \mathrm{N_B} = 1 \quad ? \quad (-1)$$

[Solve for $\mathrm{N_B}$]

$$\mathrm{N_6} = 1 - \mathrm{N_B}$$

$$\mathrm{N_7} = 1$$

[Substitute]

$$H_7 * 1 = H_6 * N_6 + H_B * (1 - \mathrm{N_6})$$

[Plug in numbers from CHE 301 Calc]

$$12549.17 = 49725N_B + 3154(1 - \mathrm{N_B})$$

[Expand]

$$12549.17 = 49725N_B + 3154 - 3154\mathrm{N_B}$$

[Combine like-terms]

$$9395.1 = 46571N_B$$

[Solve for N_B}

$$N_B = .20173$$

$$N_6 = .7982$$

[Finding the molar flow rate of N_6]

$$N_6 = .7982 * N_2$$

$$N_6 = 10.5225 \frac{kmol}{s}$$

[Finding the molar flow rate of N_B]

$$N_B = .20173 * N_2$$

$$N_B = 2.6595 \frac{kmol}{s}$$

[Finding the flow rate of N_B in MMSCFD]

$$N_B = .20173 \, * 1000 \, MMSCFD$$

$$\boldsymbol{N_B = 201.73 \, MMSCFD}$$

(-1) ?

Stream 8 Real: Low pressure pump would increase the pressure to 80 bar keeping temperature at Psat. The Enthalpy was calculated with the efficiency of 80% of the pump. It was calculated by finding the difference in enthalpy between S7 Ideal and S8 ideal divided by the efficiency plus the enthalpy of S7 Ideal. The work was the difference in enthalpy from S8 and S7 multiplied by the molar flow rate.

[Calculation for S8 Real]

$$\text{S8 real} = \frac{(\text{S8 Ideal} - \text{S7 Ideal})}{.88} + \text{S7 Ideal} \quad = ? \quad (-1)$$

[Work from Real High-Pressure pump]

$$(-1) \; ? \quad (\text{S6} - \text{S5})\text{N}_6 = 1241.52 \frac{\text{KJ}}{\text{s}} \quad \times \quad (-2)$$

2. The power generated by the cycle would be the sum of all the power from the turbines and the pump

[To find the total work]

$$W_{total} = \sum W$$

[Expand]

$$W_{total} = W_{HPT} + W_{LPT} + W_{LPP} + W_{HPP}$$

[Plug in numbers]

$$W_{total} = -126691.12 - 126862.16 + 259.20 + 1241.52$$

[Answer in Mega-Watts]

$$\boldsymbol{W_{total} = -252052.54 \frac{KJ}{s} = -252.052 \, MW}$$

3.The efficiency of the cycle

[Efficiency equation]

$$Abs\left(\frac{W_{total}}{Q_h}\right) * 100 = \text{Efficiency}$$

[Solving for Q_h]

$$Q_h = (H_1 - H_8) * N_8$$

[Plug in values]

$$Q_h = (60302.45 - 12454.99) * 13.182$$

[Final Value]

$$Q_h = 630725.11\frac{KJ}{s}$$

[Plug in the values that we calculated for Q_h and W_{total}]

$$abs\left(\frac{-252052.54}{630725.11}\right) * 100 = Efficiency$$

[Final Efficiency]

$$\boldsymbol{Eficiency = 39.96\%}$$

4. Finding the water in the condenser stream water with a rise of 11 degrees C. First we need to find out how much heat was taken out from turn S3 quality from 1 to S4 quality to 0.

[The equation used to calculate for mass flow rate]

$$Q_c = MC\Delta\mathrm{T}$$

[Solve for Q_c]

$$Q_c = (H_5 - H_4) * N_4$$

[Using the CHE 301 Calc. solve for Q_c]

$$Q_c = 376779.50\frac{KJ}{s}$$

[Plug in Q_c]

$$376779.50\frac{KJ}{s} = M * \left(\underbrace{4.18\frac{KJ}{Kg\ K°}}_{\text{at what temp?}}\right) * 11K°$$

[Solve for M]

$$M = 9878.87\frac{Kg}{s} = 9878873\frac{g}{s}$$

[Turn it into a volumetric flow rate]

$$9878873\frac{g}{s}*1\frac{cm^3}{g} = 9878873\frac{cm^3}{s}$$

[Turn it into $\frac{m^3}{s}$]

$$9878873\frac{cm^3}{s}*\frac{1m^3}{100^3cm^3} = 8.1626\frac{m^3}{s}$$

[Turn in into $\frac{gal}{min}$]

$$8.1626\frac{m^3}{s}*60\frac{s}{min}*264.172\frac{gal}{m^3} = \mathbf{129379.848\frac{gal}{min}}$$

5. Finding how much coal in tons per day is used in the plant

[Energy density of coal found online] reference ? (-2)

$$Coal = 6150\frac{KW*h}{ton}$$

[Take the inverse of the energy density]

$$\frac{1}{6150}\frac{Ton}{KW*h}$$

[Turn it into days instead of hours]

$$\frac{1}{6150}\frac{ton}{KW*h}*\frac{24h}{1\,Day} = \frac{24}{6150}\frac{Ton}{KW*Day}$$

[Multiply it by the Q_h]

$$\frac{24}{6150}\frac{Ton}{KW*Day}*630735\,KW = \mathbf{2461.36\frac{Ton}{Day}}$$

6. Finding the water draw from environment and discharge as waste

[Ratio of Draw and makeup stream]

$$\frac{N_D}{N_m} = 5$$ why ? (-5)

[solve for N_p]

$$N_p = 5N_D$$ what are N_p and N_D ? (-1)

[Mass balance]

$$M = D + E$$

[Substitute]

$$5D = D + E$$

[Simplify]

$$4D = E$$

[Equation to solve for E]

$$E = \frac{C_p * M * \Delta T}{H_{Vap}}$$ why? (-1)

[Plug in numbers]

$$E = \frac{4.186 * 11 * 129379}{\underbrace{2405.3}}$$ at what temperature? (-2)

$$E = 2476.79 \frac{Gal}{min}$$

[Solve for D]

$$D = \frac{E}{4} = \frac{2476.79}{4} = 619.19 \frac{Gal}{min}$$

$$\boldsymbol{D = 619.19 \frac{Gal}{min}}$$

[Solve for M]

$$5D = M = 5 * 619.19 = 3095.98 \frac{Gal}{min}$$

$$\boldsymbol{M = 3095.98 \frac{Gal}{min}}$$